

Simplicial Convolutions

[Yang et al. 2022 IEEE TSP]

(Helmholtz) Hodge decomposition

$$\text{Edge flow} = \begin{array}{c} \text{Curl flow} \\ \text{Div-free, solenoidal} \end{array} + \begin{array}{c} \text{Gradient flow} \\ \text{Curl-free, irrotational} \end{array} + \begin{array}{c} \text{Harmonic flow} \\ \text{Div- and curl-free} \end{array}$$

Edge flows behave like vector fields

